

Fundamentals of Operating System Design

Module 7: Processor Mode Bit and CPU Protection

Prashant Jadav
DEPSTAR-CHARUSAT

Learning Outcomes

- Explain the processor mode bit.
- Describe CPU protection mechanisms.
- Differentiate trap, interrupt and exception.
- Explain how the processor switches between modes.

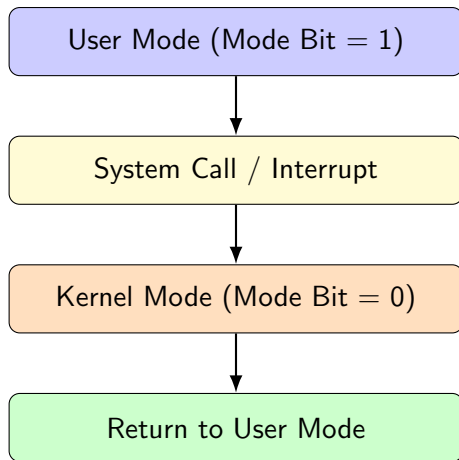
Processor Mode Bit

Definition

Modern processors maintain a mode bit (privilege bit) that indicates whether the CPU is executing in User Mode or Kernel Mode.

Mode Bit	CPU Mode
0	Kernel Mode (Privileged)
1	User Mode (Unprivileged)

How the Processor Mode Bit Changes



Events That Change CPU Mode

- System Call
- Hardware Interrupt
- Software Interrupt
- Exception
- Reset

Trap, Interrupt and Exception

Event	Description
Trap	Intentional transfer to the kernel (e.g., system call).
Interrupt	External event generated by hardware.
Exception	Internal event caused during instruction execution.

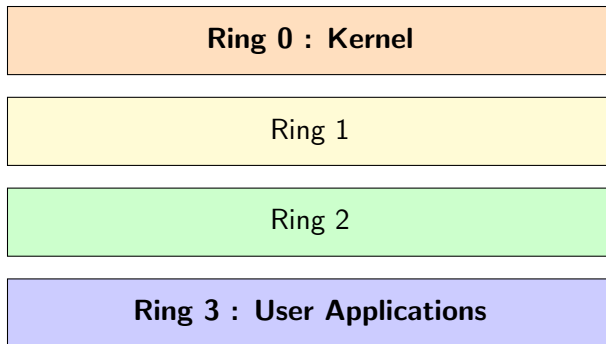
Examples

Event	Example
Trap	write(), read(), fork()
Interrupt	Keyboard press, Timer interrupt
Exception	Divide by zero, Invalid instruction

CPU Protection

- Protect operating system memory.
- Prevent unauthorized hardware access.
- Isolate user programs.
- Protect I/O devices.
- Ensure system stability.

CPU Protection Rings



Key Point

Privilege decreases from Ring 0 to Ring 3. Only Ring 0 has unrestricted access to hardware.

Execution Sequence

- 1 User program executes.
- 2 System call instruction executed.
- 3 CPU changes mode bit.
- 4 Kernel executes requested service.
- 5 Return instruction restores user mode.

Summary

- CPU uses a mode bit for protection.
- Kernel mode has full privileges.
- User mode has restricted privileges.
- Traps, interrupts and exceptions transfer control to the kernel.
- CPU protection improves reliability and security.

Quick Review

- ① What is the processor mode bit?
- ② What does mode bit value 0 represent?
- ③ Differentiate trap, interrupt and exception.
- ④ Explain CPU protection.